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[Books.php](#) – link to index on each header

PHP:

1. Use functions.php.
2. Connect to session if logged in, otherwise redirects to login.
3. Connect to the database and retrieves all book's categories.
4. Retrieve books table from the database.
5. Wait for post request with isbn value to reserve a book.
6. Call reserveBook(\$book_isbn, \$username) – functions.php

HTML:

1. Include header and footer
2. Output message, if set.
3. Print each book and its values.
4. Input through form to reserve books.

Db.sql

SQL:

1. Create tables: users, categories, books, reservedBooks.
2. Populate tables with given data.

Header.php

PHP:

1. Start session if not started yet.

HTML:

1. Output header navigation menu.
2. Welcome user.

Footer.php

PHP:

1. Start session if not started yet.

HTML:

1. Output footer content.

Functions.php

PHP:

connectDb(): mysqli

1. connect to the database
2. check connection with isConnected():

isConnected(\$connection): bool

1. if there is a connection error, die

registerUser(\$username, \$password, \$mobile): string - message

1. connect to database with connectDb() func
2. prepare and sanitize user input
3. check if the user exists with doesUserExists(\$username) func
4. insert sanitized data in db
5. check if there is an error
6. fill in session values with users data with fillInSession(\$username)
7. close connection with database

loginUser(\$username, \$password): string -message

1. connect to database with connectDb() func
2. prepare and sanitize user input
3. check password with checkPassword(\$username, \$password) func
4. if password is right, fill in session with fillInSession(\$username)func
5. close db connection

fillInSession(\$username): string – message

1. connect to db with connectDb() func
2. select all user data via sql query
3. if there is no result, return error message
4. if user was not found, return error message
5. fetch result into row
6. set all variables for session
7. close db connection

doesUserExists(\$username): bool

1. connect to db
2. put name to lower chars
3. select user from table with the given name
4. if result is false, die
5. close db connection

checkPassword(\$username, \$password): string

1. connect to db
2. get user via sql query
3. if there is no result, die
4. if user is not found, return error message
5. fetch result into row
6. if password does not match, return error message
7. close db connection

reserveBook(\$book_isbn, \$username)

1. connect to db
2. sanitize user input
3. check if the book is not reserved
4. try to reserve the book
5. update book's status in the db

6. close db connection

[Index.php](#)

PHP:

1. Start session if required
2. Check if user is not logged in and redirect to login page

HTML:

1. Include head and footer
2. Output quick action cards menu
3. Count and output total books, total categories, all reserved books by the user, all available books in database
4. Output stats of the library and user

[Login.php](#)

PHP:

1. Start session if required
2. Check if user is not logged in and redirect to login page
3. Wait for form to be filled and get values through post method
4. Check that input is not empty
5. Call loginUser(\$username, \$password) – defined in [functions.php](#)
6. If login successful, redirect to [index.php](#)
7. Else write to error message variable

HTML:

1. Output error message if it is not empty
2. Print login form
3. If the username was set, put it back in the form
4. Print link to register

Logout.php

PHP:

1. If session is not started, start it
2. Destroy session
3. Redirect to login page

Profile.php

PHP:

1. Start session if required
2. Check if user is not logged in and redirect to login page
3. Wait for form to be submitted
4. Connect to db
5. Sanitize the user input data
6. Update database values for the user with provided
7. Check if the query was successful
8. Update the rest of the session values for the user

HTML:

1. Include header and footer
2. Output error message if not empty
3. Fill in the form with the sanitized values that are in session

Register.php

PHP:

1. Start session if required
2. Check if user is logged in and redirect to index page
3. Wait for form to be submitted
4. Check if the passwords match
5. Call registerUser(\$username, \$password) – defined in [functions.php](#)
6. If the error message is empty, redirect to index.php

HTML:

1. If error message is not empty output error
2. Print form and put post elements if already set previously
3. Make sure that password has at least one number, letter and capital letter
4. Output hint for the password required pattern
5. Print link to login page

Reserved.php

PHP:

1. Start session if required
2. Check if user is not logged in and redirect to login page
3. Connect to db
4. Select users reserved books from db
5. Fetch all rows into books array
6. Wait for form to be submitted, remove reservation button pressed
7. Try to remove form reservedBooks table
8. Try to update book reserved status
9. Refresh the page

HTML:

1. Include header and footer
2. If error message is not empty output it
3. If there any books reserved, output it

[Search.php](#)

PHP:

1. Check if user is not logged in and redirect to login page
2. Connect to db
3. Select book's categories
4. Wait for search button to be pressed
5. Sanitize user input
6. Compile a sql query to select a book with user input
7. If value is not empty, user it for search
8. If any books are found, go through them and input into an array
9. Call reserveBook(\$isbn, \$username) – function.php

HTML:

1. Output reserve message if not empty
2. Output form for search
3. Autofill form if was filled before
4. Output drop down menu for categories
5. Output search message
6. If search is not empty, go through the books array and output each
7. Print books values
8. If the book is available, show the reserve button and hide the form

Screenshots – link to index

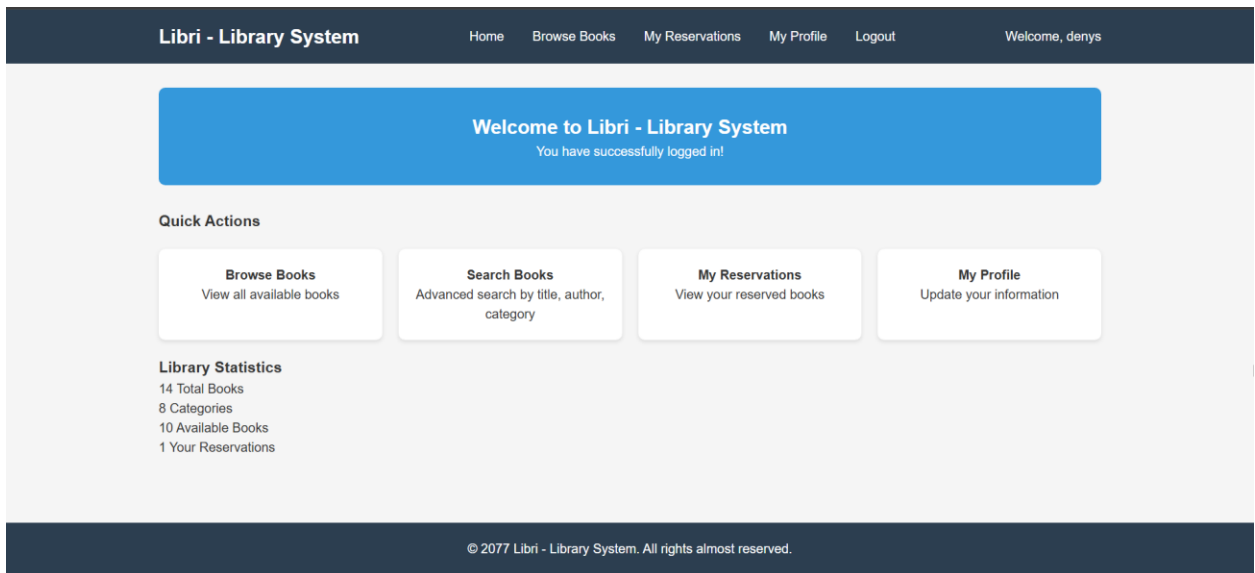
Header.php



Footer.php



Index.php



Books.php

Libri - Library System

[Home](#)[Browse Books](#)[My Reservations](#)[My Profile](#)[Logout](#)

Welcome, denys

Browse All Books

Advanced Search →

A guide to nutrition

Available

Author: John Thorpe

Category: Health

ISBN: 2318-923849

Year: 1997

Edition: 2

Reserve This Book

Business Strategy

Reserved

Author: Joe Peppard

Category: Business

ISBN: 237-34823

Year: 2002

Edition: 2

This book is already reserved

computers for idiots

Available

Author: Susan O'Neill

Category: Technology

ISBN: 8218-308

Year: 1998

Edition: 5

Reserve This Book

Computers in Business

Reserved

Author: Alicia Oneill

Category: Biography

ISBN: 003-403002

Cooking for children

Available

Author: Anabelle Sharpe

Category: Cookery

ISBN: 2083-3404

DaVinci Code

Available

Author: Dan Brown

Category: Fiction

ISBN: 0823-2403-0

Reserved.php

Libri - Library System

[Home](#)[Browse Books](#)[My Reservations](#)[My Profile](#)[Logout](#)

Welcome, denys

My Reservations

Title	Author	ISBN	Category	Reservation Date	Actions
A guide to nutrition	John Thorpe	2318-923849	Health	05-Dec-2025	Remove
Business Strategy	Joe Peppard	237-34823	Business	04-Dec-2025	Remove

You have 2 reserved book(s).

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Profile.php

Libri - Library System[Home](#)[Browse Books](#)[My Reservations](#)[My Profile](#)[Logout](#)Welcome, denys

My Profile

First Name:

Last Name:

Address Line 1:

Address Line 2:

City:

Telephone:

Username: denys

Login.php

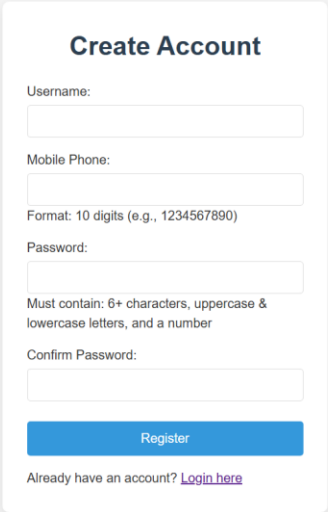
Login to Libri

Username:

Password:

Don't have an account? [Register here](#)

Register.php



The image shows a 'Create Account' registration form. It has a title 'Create Account' at the top. Below the title are four input fields: 'Username:', 'Mobile Phone:', 'Password:', and 'Confirm Password:'. The 'Mobile Phone:' field has a note below it: 'Format: 10 digits (e.g., 1234567890)'. The 'Password:' field has a note below it: 'Must contain: 6+ characters, uppercase & lowercase letters, and a number'. At the bottom of the form is a blue 'Register' button. Below the button is a link: 'Already have an account? [Login here](#)'.

[Source code](#) – link to index

1. [Books.php](#)

```
<?php
require_once 'functions.php';

// start session if required
if (session_status() === PHP_SESSION_NONE) {
    session_start();
}

// if session is on but user is not logged_in
if(!isset($_SESSION["logged_in"]) || $_SESSION["logged_in"] === FALSE) {
    // redirect to login page and exit
    session_destroy();
    header("Location: login.php");
    exit();
}

$conn = connectDb();
$books = [];
```

```

$categories = [];

// run sql query to retrieve books categories
$category_result = $conn->query("SELECT * FROM categories ORDER BY
description");
while($row = $category_result->fetch_assoc()) {
    $categories[] = $row;
}

// run the sql query and retrieve books table
$sql = "SELECT b.*, c.description as category_name FROM books b
        LEFT JOIN categories c ON b.category = c.id
        ORDER BY b.title";
$book_result = $conn->query($sql);
while($row = $book_result->fetch_assoc()) {
    $books[] = $row;
}

// handle reservation
if($_SERVER["REQUEST_METHOD"] == "POST" && isset($_POST['reserve_isbn'])) {
    $message = reserveBook($_POST["reserve_isbn"], $_SESSION["username"]);
}

$conn->close();
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Browse Books - Libri</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <?php include 'header.php'; ?>

    <main class="main-content">
        <div class="content-container">
            <h1 class="page-title">Browse All Books</h1>

            <div class="search-container">
                <button class="search-button"><a href="search.php" class="search-
link">Advanced Search →</a></button>
            </div>

```

```

        <!-- output message -->
        <?php if(isset($message)) {?>
            <?php echo $message; ?>
        <?php } ?>

        <div class="books-section">
            <div class="books-grid">
                <!-- go through each book and echo it's values -->
                <?php foreach($books as $book) {?>
                    <div class="book-card <?php echo $book['reserved'] ?
'reserved' : 'available'; ?>">
                        <div class="book-header">
                            <!-- echo header of the book -->
                            <h3 class="book-title"><?php echo $book['title'];
?></h3>

                            <div class="book-status">
                                <span class="status-badge <?php echo
$book['reserved'] ? 'status-reserved' : 'status-available'; ?>">
                                    <?php echo $book['reserved'] ? 'Reserved'
: 'Available'; ?>
                                </span>
                            </div>
                        </div>

                        <!-- echo values -->
                        <div class="book-details">
                            <p class="book-author"><strong>Author:</strong>
<?php echo $book['author']; ?></p>
                            <p class="book-
category"><strong>Category:</strong> <?php echo $book['category_name']; ?></p>
                            <p class="book-isbn"><strong>ISBN:</strong> <?php
echo $book['isbn']; ?></p>
                            <p class="book-year"><strong>Year:</strong> <?php
echo $book['year']; ?></p>
                            <p class="book-edition"><strong>Edition:</strong>
<?php echo $book['edition']; ?></p>
                        </div>

                        <!-- show form to reserve the book if available -->
                        <?php if(!$book['reserved']) {?>
                            <form method="post" class="reserve-form">
                                <input type="hidden" name="reserve_isbn"
value="<?php echo $book['isbn']; ?>">

```

```

        <button type="submit" class="reserve-
btn">Reserve This Book</button>
    </form>
    <!-- otherwise show that the book is reserved -->
    <?php } else {?>
        <div class="reserved-info">
            <p class="reserved-text">This book is already
reserved</p>
        </div>
    <?php }?>
</div>
<?php }?>
</div>
</div>
</div>
</main>

    <?php include 'footer.php'; ?>
</body>
</html>

```

2. Db.sql

```

-- creating tables

-- users table
CREATE TABLE IF NOT EXISTS users (
    username VARCHAR(30) PRIMARY KEY,
    password VARCHAR(30) NOT NULL,
    fname VARCHAR(50),
    sname VARCHAR(50),
    address1 VARCHAR(100),
    address2 VARCHAR(100),
    city VARCHAR(30),
    tel VARCHAR(20),
    mobile VARCHAR(20) NOT NULL
);

-- book's categories table
CREATE TABLE IF NOT EXISTS categories (
    id VARCHAR(3) PRIMARY KEY,
    description VARCHAR(50) UNIQUE NOT NULL
);

```



```

-- books table
CREATE TABLE IF NOT EXISTS books (
    isbn VARCHAR(20) PRIMARY KEY,
    title VARCHAR(50) NOT NULL,
    author VARCHAR(100),
    edition INT,
    year INT,
    category VARCHAR(3) REFERENCES categories(id),
    resell BOOLEAN NOT NULL
);

-- reserved books table
CREATE TABLE IF NOT EXISTS reservedBooks (
    isbn VARCHAR(20) REFERENCES books(isbn),
    username VARCHAR(30) REFERENCES users(username),
    date VARCHAR(20) NOT NULL
);

-- populating database

-- insert sample data into categories table
INSERT INTO categories (id, description) VALUES
('001', 'Health'),
('002', 'Business'),
('003', 'Biography'),
('004', 'Technology'),
('005', 'Travel'),
('006', 'Self-Help'),
('007', 'Cookery'),
('008', 'Fiction');

-- insert sample data into users table
INSERT INTO users (username, password, fname, sname, address1, address2, city,
tel, mobile) VALUES
('alanjmckenna', 't1234s', 'Alan', 'McKenna', '38 Cranley Road', 'Fairview',
'Dublin', '9998377', '856625567'),
('joecrotty', 'kj7899', 'Joseph', 'Crotty', 'Apt 5 Clyde Road', 'Donnybrook',
'Dublin', '8887889', '876654456'),
('tommy100', '123456', 'Tom', 'Behan', '14 Hyde Road', 'Dalkey', 'Dublin',
'9983747', '876738782');

-- insert sample data into books table
INSERT INTO books (isbn, title, author, edition, year, category, resell) VALUES
('093-403992', 'Computers in Business', 'Alicia Oneill', 3, 1997, '003', false),

```

```
(
'23472-8729', 'Exploring Peru', 'Stephanie Birchi', 4, 2005, '005', false),
('237-34823', 'Business Strategy', 'Joe Peppard', 2, 2002, '002', false),
('2318-923849', 'A guide to nutrition', 'John Thorpe', 2, 1997, '001', false),
('2983-3494', 'Cooking for children', 'Anabelle Sharpe', 1, 2003, '007', false),
('8218-308', 'computers for idiots', 'Susan O'Neill', 5, 1998, '004', false),
('9823-23984', 'My life in picture', 'Kevin Graham', 8, 2004, '001', false),
('9823-2403-0', 'DaVinci Code', 'Dan Brown', 1, 2003, '008', false),
('98234-029384', 'My ranch in Texas', 'George Bush', 1, 2005, '001', true),
('9823-98345', 'How to cook Italian food', 'Jamie Oliver', 2, 2005, '007', true),
('9823-98487', 'Optimising your business', 'Cleo Blair', 1, 2001, '002', false),
('988745-234', 'Tara Road', 'Maeve Binchy', 4, 2002, '008', false),
('993-004-00', 'My life in bits', 'John Smith', 1, 2001, '001', false),
('9987-0039882', 'Shooting History', 'Jon Snow', 1, 2003, '001', false);

-- insert sample data into reservations table
INSERT INTO reservedBooks (isbn, username, date) VALUES
('98234-029384', 'joecrotty', '11-Oct-2008'),
('9823-98345', 'tommy100', '11-Oct-2008');
```

3. Header.php

```
<?php
    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Libri - Library System</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header class="site-header">
        <div class="header-container">
            <h1 class="site-title">Libri - Library System</h1>
            <nav class="main-nav">
                <ul class="nav-list">
                    <li class="nav-item"><a href="index.php" class="nav-
link">Home</a></li>
```

```

        <li class="nav-item"><a href="books.php" class="nav-
link">Browse Books</a></li>
        <li class="nav-item"><a href="reserved.php" class="nav-
link">My Reservations</a></li>
        <li class="nav-item"><a href="profile.php" class="nav-
link">My Profile</a></li>
        <li class="nav-item"><a href="logout.php" class="nav-
link">Logout</a></li>
    </ul>
</nav>
<div class="user-info">
    <span class="welcome-message">Welcome, <?php echo
$_SESSION['username'] ?? 'Guest'; ?></span>
</div>
</div>
</header>

```

4. Footer.php

```

<?php
    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <footer class="site-footer">
        <div class="footer-container">
            <p class="copyright">&copy; 2077 Libri - Library System. All rights
almost reserved.</p>
        </div>
    </footer>
</body>
</html>

```

5. Functions.php

```
<?php
function connectDb(): mysqli {
    $servername = "localhost";
    $host_name = "root";
    $host_password = "";
    $dbname = "libri";

    // create connection to my database
    $conn = new mysqli($servername, $host_name, $host_password, $dbname);
    // check connection
    isConnected($conn);

    return $conn;
}

// check connection
function isConnected(mysqli $conn): bool {
    // if connection failed, output error message and exit
    if($conn->connect_error) {
        die("Connection failed: " . $conn->connect_error);
    }
    return !$conn->connect_error;
}

// register user
function registerUser($uname, $password, $mobile): string {
    $conn = connectDb();

    // error message
    $message = "";

    // prepare sanitized input
    $safe_uname = $conn->real_escape_string($uname);
    $safe_password = $conn->real_escape_string($password);
    $safe_mobile = $conn->real_escape_string($mobile);

    // if a user with this name is already exists
    if(doesUserExist($safe_uname)) {
        // give error message and exit
        $message = "Username is already taken";
    }
}
```

```

        $conn->close();
        return $message;
    }

    // insert sanitized data
    $sql = "INSERT INTO users (username, password, mobile) VALUES
    ('$safe_uname', '$safe_password', '$safe_mobile')";

    // execute sql query
    $result = $conn->query($sql);

    // if there is an error
    if(!$result) {
        $message = "There was something wrong with your data";
        $conn->close();
        return $message;
    }

    $message = fillInSession($safe_uname);

    $conn->close();

    // return the message regardless of it's content
    return $message;
}

function loginUser($uname, $password): string {
    $conn = connectDb();

    // prepare case insensitive and sanitized input
    $uname_lower = strtolower($uname);
    $safe_uname = $conn->real_escape_string($uname_lower);

    // get message from function
    $message = checkPassword($safe_uname, $password);

    // if message is empty, fill in the session info on the user
    if(!$message) {
        $message = fillInSession($safe_uname);
    }

    $conn->close();

    // return the message regardless of it's content
    return $message;
}

```

```

}

function fillInSession($safe_uname): string {
    $conn = connectDb();

    // error message
    $message = "";

    // sql query to get all user data
    $sql = "SELECT username, password, fname, sname, address1, address2,
city, tel, mobile
        FROM users WHERE username = '$safe_uname'";

    // query and store in result
    $result = $conn->query($sql);

    // if there is no result
    if(!$result) {
        $message = "Failed to check if user exists: " . $conn->error;
        $conn->close();
        return $message;
    }

    // user was not found
    if(!($result->num_rows > 0)) {
        $message = "User not found! from session";
        $conn->close();
        return $message;
    }

    $row = $result->fetch_assoc();

    // set all variables for session
    $_SESSION['username'] = $row['username'];
    $_SESSION['password'] = $row['password'];
    $_SESSION['fname'] = $row['fname'];
    $_SESSION['sname'] = $row['sname'];
    $_SESSION['address1'] = $row['address1'];
    $_SESSION['address2'] = $row['address2'];
    $_SESSION['city'] = $row['city'];
    $_SESSION['tel'] = $row['tel'];
    $_SESSION['mobile'] = $row['mobile'];
    $_SESSION['logged_in'] = true;

    $conn->close();
}

```

```

        return $message;
    }

    function doesUserExist($safe_uname): bool {
        $conn = connectDb();

        // prepare case insensitive and sanitized input
        $uname_lower = strtolower($safe_uname);

        // execute sql query
        $sql = "SELECT username FROM users WHERE LOWER(username) =
'$uname_lower'";
        $result = $conn->query($sql);

        // if result is false, die
        if (!$result) {
            $conn->close();
            die("Failed to check if user exists: " . $conn->error);
        }

        $conn->close();

        // return true if found user
        return $result->num_rows > 0;
    }

    function checkPassword($safe_uname, $password): string {
        $conn = connectDb();

        // error message init
        $message = "";

        // look for provided username
        $sql = "SELECT username, password FROM users WHERE LOWER(username) =
'$safe_uname'";
        $result = $conn->query($sql);

        // if there is no result
        if (!$result) {
            $conn->close();
            die("Failed to check if user exists: " . $conn->error);
        }

        // if there is no rows / aka user is not found

```

```

        if(!($result->num_rows > 0)) {
            $message = "User not found!";
            $conn->close();
            return $message;
        }

        // take a row
        $row = $result->fetch_assoc();

        // check password
        if($row["password"] != $password) {
            $message = "Incorrect password!";
            $conn->close();
            return $message;
        }

        // if all was right and password matched, message should be empty
        $conn->close();
        return $message;
    }

    function reserveBook($book_isbn, $uname): string {
        $conn = connectDb();
        $message = "";

        // sanitize input in case user could input value somehow
        $isbn = $conn->real_escape_string($book_isbn);
        $username = $conn->real_escape_string($uname);
        $date = date('d-M-Y');

        // check if the book is reserved
        $check_book = $conn->query("SELECT reserved FROM books WHERE isbn = '$isbn'");
        $book_data = $check_book->fetch_assoc();

        // if book is not reserved
        if($book_data && $book_data['reserved'] == false) {
            try {
                // reserve the book
                $insert_sql = "INSERT INTO reservedBooks (isbn, username, date)
VALUES ('$isbn', '$username', '$date')";
                $conn->query($insert_sql);

                // update books status in database

```



```

        $update_sql = "UPDATE books SET reserved = true WHERE isbn =
'$isbn'";

        $conn->query($update_sql);

        // output success message
        $message = '<div class="success-message">Book reserved
successfully!</div>';

    } catch (Exception $e) {
        // if exception is caught, output error message
        $message = '<div class="error-message">Failed to reserve book: '
. $e->getMessage() . '</div>';
    }
    } else {
        // error, the book is reserved
        $message = '<div class="error-message">This book is already reserved
by someone else!</div>';
    }

    $conn->close();
    return $message;
}
?>

```

6. Index.php

```

<?php
    require_once 'functions.php';

    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }

    // if session is on but user is not logged_in
    if(!isset($_SESSION["logged_in"]) || $_SESSION["logged_in"] === FALSE) {
        // redirect to login page and exit
        session_destroy();
        header("Location: login.php");
        exit();
    }
?>
<!DOCTYPE html>
<html lang="en">

```

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Home - Libri</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <?php include 'header.php'; ?>

  <main class="main-content">
    <div class="content-container">
      <section class="welcome-section">
        <h2 class="section-title">Welcome to Libri - Library System</h2>
        <p class="welcome-text">You have successfully logged in!</p>
      </section>

      <section class="quick-actions">
        <h3 class="actions-title">Quick Actions</h3>
        <div class="actions-grid">
          <a href="books.php" class="action-card">
            <h4 class="card-title">Browse Books</h4>
            <p class="card-text">View all available books</p>
          </a>

          <a href="search.php" class="action-card">
            <h4 class="card-title">Search Books</h4>
            <p class="card-text">Advanced search by title, author,
category</p>
          </a>

          <a href="reserved.php" class="action-card">
            <h4 class="card-title">My Reservations</h4>
            <p class="card-text">View your reserved books</p>
          </a>

          <a href="profile.php" class="action-card">
            <h4 class="card-title">My Profile</h4>
            <p class="card-text">Update your information</p>
          </a>
        </div>
      </section>

      <section class="stats-section">
        <h3 class="stats-title">Library Statistics</h3>
        <div class="stats-container">
```

```

        <?php
            $conn = connectDb();
            // count total books in db
            $total_books = $conn->query("SELECT COUNT(*) as total
FROM books")->fetch_assoc()['total'];
            // count total categories in db
            $total_categories = $conn->query("SELECT COUNT(*) as
total FROM categories")->fetch_assoc()['total'];
            // count all reserved books by the user in db
            $reserved_books = $conn->query("SELECT COUNT(*) as total
FROM reservedBooks WHERE username = '".$_SESSION['username']."'")-
>fetch_assoc()['total'];
            // count all available books in db
            $available_books = $conn->query("SELECT COUNT(*) as total
FROM books WHERE reserved = false")->fetch_assoc()['total'];
            $conn->close();
        ?>
        <div class="stat-item">
            <span class="stat-number"><?php echo $total_books;
?></span>

            <span class="stat-label">Total Books</span>
        </div>
        <div class="stat-item">
            <span class="stat-number"><?php echo $total_categories;
?></span>

            <span class="stat-label">Categories</span>
        </div>
        <div class="stat-item">
            <span class="stat-number"><?php echo $available_books;
?></span>

            <span class="stat-label">Available Books</span>
        </div>
        <div class="stat-item">
            <span class="stat-number"><?php echo $reserved_books;
?></span>

            <span class="stat-label">Your Reservations</span>
        </div>
    </div>
</section>
</div>
</main>

    <?php include 'footer.php'; ?>
</body>
</html>

```

7. Login.php

```
<?php
    require_once 'functions.php';

    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }

    // if logged in
    if(isset($_SESSION["logged_in"]) && $_SESSION["logged_in"] === TRUE) {
        // redirect to main page
        header("Location: index.php");
        exit();
    }

    $message = "";

    if($_SERVER["REQUEST_METHOD"] == "POST") {
        $username = $_POST["username"];
        $password = $_POST["password"];

        if(empty($username) || empty($password)) {
            $message = '<div class="error-message">Please fill in all
fields</div>';
        } else {
            $message = loginUser($username, $password);
            if(empty($message)) {
                header("Location: index.php");
                exit();
            } else {
                $message = '<div class="error-message">' . $message . '</div>';
            }
        }
    }
}

?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login - Libri</title>
    <link rel="stylesheet" href="style.css">
```

```

</head>
<body class="auth-body">
  <div class="auth-form-container">
    <h1 class="auth-title">Login to Libri</h1>

    <?php
      // if message is not empty, echo it
      if($message) {
        echo $message;
      }
    ?>

    <!-- login form -->
    <form method="post" class="login-form">
      <div class="form-group">
        <label for="username" class="form-label">Username:</label>
        <!-- if the username was set, put it in back -->
        <input type="text" id="username" name="username" class="form-
input"
          value="<?php echo isset($_POST['username']) ?
htmlentities($_POST['username']) : ''; ?>"
          required>
      </div>

      <div class="form-group">
        <label for="password" class="form-label">Password:</label>
        <input type="password" id="password" name="password" class="form-
input" required>
      </div>

      <div class="form-group">
        <input type="submit" value="Login" class="submit-btn">
      </div>
    </form>

    <div class="auth-link">
      Don't have an account? <a href="register.php">Register here</a>
    </div>
  </div>
</body>
</html>

```

8. Logout.php

```

<?php
    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }

    session_destroy();
    header("Location: login.php");
    exit();
?>

```

9. Profile.php

```

<?php
    require_once 'functions.php';

    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }

    // if session is on but user is not logged_in
    if(!isset($_SESSION["logged_in"]) || $_SESSION["logged_in"] === FALSE) {
        // redirect to login page and exit
        session_destroy();
        header("Location: login.php");
        exit();
    }

    $message = '';

    // if the form is submitted
    if($_SERVER["REQUEST_METHOD"] == "POST") {
        $conn = connectDb();

        // sanitize input
        $fname = $conn->real_escape_string($_POST['fname']);
        $sname = $conn->real_escape_string($_POST['sname']);
        $address1 = $conn->real_escape_string($_POST['address1']);
        $address2 = $conn->real_escape_string($_POST['address2']);
        $city = $conn->real_escape_string($_POST['city']);
        $tel = $conn->real_escape_string($_POST['tel']);
        $username = $_SESSION['username'];
    }

```

```

        // update database values for the user
        $sql = "UPDATE users SET
                fname = '$fname',
                sname = '$sname',
                address1 = '$address1',
                address2 = '$address2',
                city = '$city',
                tel = '$tel'
                WHERE username = '$username'";

        // if the query was successful
        if($conn->query(query: $sql)) {
            $message = '<div class="success-message">Profile updated
successfully!</div>';
            // update the rest of session values for the user
            $_SESSION['fname'] = $fname;
            $_SESSION['sname'] = $sname;
            $_SESSION['address1'] = $address1;
            $_SESSION['address2'] = $address2;
            $_SESSION['city'] = $city;
            $_SESSION['tel'] = $tel;
        } else {
            $message = '<div class="error-message">Error updating profile: ' .
$conn->error . '</div>';
        }

        $conn->close();
    }
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Profile - Libri</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <?php include 'header.php'; ?>

    <main class="main-content">
        <div class="content-container">
            <h1 class="page-title">My Profile</h1>

```

```

    <?php
        // if the message is not empty, output it
        if($message) {
            echo $message;
        }
    ?>

    <!-- output form to finish registration -->
    <div class="profile-section">
        <form method="post" class="profile-form">
            <div class="form-row">
                <div class="form-group">
                    <label for="fname" class="form-label">First
Name:</label>

                    <!-- if the first name is in session, sanitize it and
put it in -->

                    <input type="text" id="fname" name="fname"
class="form-input"
                        value="<?php echo
htmlentities($_SESSION['fname'] ?? ''); ?>">
                    </div>

                    <div class="form-group">
                        <label for="sname" class="form-label">Last
Name:</label>

                        <!-- if the second name is in session, sanitize it
and put it in -->

                        <input type="text" id="sname" name="sname"
class="form-input"
                            value="<?php echo
htmlentities($_SESSION['sname'] ?? ''); ?>">
                        </div>
                    </div>

                    <div class="form-group">
                        <label for="address1" class="form-label">Address Line
1:</label>

                        <!-- if the address1 is in session, sanitize it and put
it in -->

                        <input type="text" id="address1" name="address1"
class="form-input"
                            value="<?php echo
htmlentities($_SESSION['address1'] ?? ''); ?>">
                        </div>

```



```

        <div class="form-group">
            <label for="address2" class="form-label">Address Line
2:</label>
            <!-- if the address2 is in session, sanitize it and put
it in -->
            <input type="text" id="address2" name="address2"
class="form-input"
                value="<?php echo
htmlentities($_SESSION['address2'] ?? ''); ?>">
            </div>

        <div class="form-row">
            <div class="form-group">
                <label for="city" class="form-label">City:</label>
                <!-- if the city is in session, sanitize it and put
it in -->
                <input type="text" id="city" name="city" class="form-
input"
                    value="<?php echo
htmlentities($_SESSION['city'] ?? ''); ?>">
                </div>

            <div class="form-group">
                <label for="tel" class="form-
label">Telephone:</label>
                <!-- if the telephone is in session, sanitize it and
put it in -->
                <input type="text" id="tel" name="tel" class="form-
input"
                    value="<?php echo
htmlentities($_SESSION['tel'] ?? ''); ?>">
                </div>
            </div>

        <div class="readonly-info">
            <div class="info-item">
                <span class="info-label">Username:</span>
                <!-- sanitize before output -->
                <span class="info-value"><?php echo
htmlentities($_SESSION['username']); ?></span>
            </div>
            <div class="info-item">
                <span class="info-label">Mobile:</span>

```

```

        <span class="info-value"><?php echo
htmlentities($_SESSION['mobile']); ?></span>
    </div>
</div>

    <div class="form-group">
        <input type="submit" value="Update Profile"
class="submit-btn">
    </div>
</form>
</div>
</div>
</main>

    <?php include 'footer.php'; ?>
</body>
</html>

```

10. Register.php

```

<?php
    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }

    // if logged in
    if(isset($_SESSION["logged_in"]) && $_SESSION["logged_in"] === TRUE) {
        // redirect to main page
        header("Location: index.php");
        exit();
    }

    $message = "";

    // if the form is submitted
    if(isset($_POST["name"]) && isset($_POST["mobile"])
    && isset($_POST["pw"]) && isset($_POST["cpw"])) {
        // if the passwords do not match
        if($_POST["pw"] != $_POST["cpw"]) {
            $message = "<div class='error-message'>Passwords do not
match!</div>";
        } else {

```

```

        require_once "functions.php";
        $message =
registerUser($_POST["name"],$_POST["pw"],$_POST["mobile"]);

        // if there is no error
        if(empty($message) && isset($_SESSION["logged_in"]) &&
$_SESSION["logged_in"] === TRUE) {
            header("Location: index.php");
            exit();
        } else {
            $message = "<div class='error-message'>$message</div>";
        }
    }
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Register - Libri</title>
    <link rel="stylesheet" href="style.css">
</head>
<body class="auth-body">
    <div class="auth-form-container">
        <h1 class="auth-title">Create Account</h1>

        <!-- error output -->
        <?php
            if($message) {
                echo $message;
            }
        ?>

        <form action="register.php" method="post" autocomplete="on"
id="registerForm">
            <div class="form-group">
                <label for="name" class="form-label">Username:</label>
                <!-- input sanitized post value if set -->
                <input type="text" id="name" name="name" class="form-input"
                    value="<?php echo isset($_POST['name']) ?
htmlentities($_POST['name']) : ''; ?>"
                    required>
            </div>

```

```

        <div class="form-group">
            <label for="mobile" class="form-label">Mobile Phone:</label>
            <!-- input sanitized post value if set -->
            <input type="tel" id="mobile" name="mobile" class="form-input"
                value="<?php echo isset($_POST['mobile']) ?
htmlentities($_POST['mobile']) : ''; ?>"
                required pattern="[\\d]{10}"
                title="Must be 10 digits">
            <div class="password-requirements">Format: 10 digits (e.g.,
1234567890)</div>
        </div>

        <div class="form-group">
            <label for="pw" class="form-label">Password:</label>
            <input type="password" id="pw" name="pw" class="form-input"
                required pattern="(?=.*\\d)(?=.*[a-z])(?=.*[A-Z]).{6,}"
                title="Must contain at least one number, one uppercase
and lowercase letter, and at least 6 characters">
            <div class="password-requirements">Must contain: 6+ characters,
uppercase & lowercase letters, and a number</div>
        </div>

        <div class="form-group">
            <label for="cpw" class="form-label">Confirm Password:</label>
            <input type="password" id="cpw" name="cpw" class="form-input"
                required pattern="(?=.*\\d)(?=.*[a-z])(?=.*[A-Z]).{6,}">
            <div id="passwordMatchMessage" class="password-mismatch"></div>
        </div>

        <div class="form-group">
            <input type="submit" value="Register" class="submit-btn"
id="submitBtn">
        </div>
    </form>

    <div class="auth-link">
        Already have an account? <a href="login.php">Login here</a>
    </div>
</div>
</body>
</html>

```

11. Reserved.php

```
<?php
    require_once 'functions.php';

    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }

    // if session is on but user is not logged_in
    if(!isset($_SESSION["logged_in"]) || $_SESSION["logged_in"] === FALSE) {
        // redirect to login page and exit
        session_destroy();
        header("Location: login.php");
        exit();
    }

    $conn = connectDb();
    $reserved_books = [];

    // select users reserved books from db
    $sql = "SELECT b.*, rb.date as reservation_date, c.description as
category_name
        FROM reservedBooks rb
        JOIN books b ON rb.isbn = b.isbn
        LEFT JOIN categories c ON b.category = c.id
        WHERE rb.username = '". $_SESSION['username'] ."'
        ORDER BY rb.date DESC";
    $result = $conn->query($sql);

    // fetch all rows to books array
    while($row = $result->fetch_assoc()) {
        $reserved_books[] = $row;
    }

    // handle remove reservation
    if($_SERVER["REQUEST_METHOD"] == "POST" && isset($_POST['remove_isbn'])) {
        $isbn = $conn->real_escape_string($_POST['remove_isbn']);
        $username = $conn->real_escape_string($_SESSION['username']);

        try {
            // remove from reservedBooks
            $delete_sql = "DELETE FROM reservedBooks WHERE isbn = '$isbn' AND
username = '$username'";
```

```

        $conn->query($delete_sql);

        // update book reserved status
        $update_sql = "UPDATE books SET reserved = false WHERE isbn =
'$isbn'";
        $conn->query($update_sql);

        // refresh the page
        header("Location: reserved.php");
        exit();

    } catch (Exception $e) {
        $error_message = '<div class="error-message">Failed to remove
reservation: ' . $e->getMessage() . '</div>';
    }
}

$conn->close();
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Reservations - Libri</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <?php include 'header.php'; ?>

    <main class="main-content">
        <div class="content-container">
            <h1 class="page-title">My Reservations</h1>

            <?php if(isset($error_message)): ?>
                <?php echo $error_message; ?>
            <?php endif; ?>

            <?php if(empty($reserved_books)): ?>
                <div class="empty-state">
                    <p class="empty-message">You have no reserved books.</p>
                    <a href="books.php" class="browse-link">Browse Books</a>
                    <a href="search.php" class="browse-link">Search Books</a>
                </div>
            </div>
        </div>
    </main>
</body>
</html>

```

```

<?php else: ?>
    <div class="reservations-table">
        <table class="data-table">
            <thead>
                <tr class="table-header">
                    <th class="table-cell">Title</th>
                    <th class="table-cell">Author</th>
                    <th class="table-cell">ISBN</th>
                    <th class="table-cell">Category</th>
                    <th class="table-cell">Reservation Date</th>
                    <th class="table-cell">Actions</th>
                </tr>
            </thead>
            <tbody>
                <?php foreach($reserved_books as $book): ?>
                    <tr class="table-row">
                        <td class="table-cell"><?php echo
htmlentities($book['title']); ?></td>
                        <td class="table-cell"><?php echo
htmlentities($book['author']); ?></td>
                        <td class="table-cell"><?php echo
htmlentities($book['isbn']); ?></td>
                        <td class="table-cell"><?php echo
htmlentities($book['category_name']); ?></td>
                        <td class="table-cell"><?php echo
htmlentities($book['reservation_date']); ?></td>
                        <td class="table-cell">
                            <form method="post" class="remove-form"
onsubmit="return confirm('Are you sure you want to remove this reservation?');">
                                <input type="hidden"
name="remove_isbn" value="<?php echo $book['isbn']; ?>">
                                <button type="submit" class="remove-
btn">Remove</button>
                            </form>
                        </td>
                    </tr>
                <?php endforeach; ?>
            </tbody>
        </table>
    </div>

    <div class="reservation-count">
        <p class="count-text">You have <?php echo
count($reserved_books); ?> reserved book(s).</p>
    </div>

```

```

        <?php endif; ?>
    </div>
</main>

    <?php include 'footer.php'; ?>
</body>
</html>

```

12. Search.php

```

<?php
    require_once 'functions.php';

    // start session if required
    if (session_status() === PHP_SESSION_NONE) {
        session_start();
    }

    // if session is on but user is not logged_in
    if(!isset($_SESSION["logged_in"]) || $_SESSION["logged_in"] === FALSE) {
        // redirect to login page and exit
        session_destroy();
        header("Location: login.php");
        exit();
    }

    $conn = connectDb();
    $categories = [];
    $search_results = [];
    $search_message = '';
    $reserve_message = '';

    // get all categories for dropdown menu
    $category_result = $conn->query("SELECT * FROM categories ORDER BY
description");
    while($row = $category_result->fetch_assoc()) {
        $categories[] = $row;
    }

    // handle search
    if($_SERVER["REQUEST_METHOD"] == "POST") {
        // sanitize the input
        $title = $conn->real_escape_string($_POST['title'] ?? '');
    }

```



```

$author = $conn->real_escape_string($_POST['author'] ?? '');
$category = $conn->real_escape_string($_POST['category'] ?? '');

// compile a select query with user input
$sql = "SELECT b.*, c.description as category_name FROM books b
        LEFT JOIN categories c ON b.category = c.id
        WHERE 1=1";

// if value is not empty, include it in the search
if(!empty($title)) {
    $sql .= " AND b.title LIKE '%$title%'";
}

if(!empty($author)) {
    $sql .= " AND b.author LIKE '%$author%'";
}

if(!empty($category)) {
    $sql .= " AND b.category = '$category'";
}

$sql .= " ORDER BY b.title";

$result = $conn->query($sql);

// if there are books found
if($result->num_rows > 0) {
    // fetch them and put them into search result
    while($row = $result->fetch_assoc()) {
        $search_results[] = $row;
    }
} else {
    $search_message = '<p class="no-results">No books found matching your
search criteria.</p>';
}

}

// handle reservation
if($_SERVER["REQUEST_METHOD"] == "POST" && isset($_POST['reserve_isbn'])) {
    $reserve_message = reserveBook($_POST["reserve_isbn"],
$_SESSION["username"]);
}

$conn->close();
?>

```

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Search Books - Libri</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <?php include 'header.php'; ?>

    <main class="main-content">
        <div class="content-container">
            <h1 class="page-title">Search Books</h1>

            <!-- output message -->
            <?php
                if($reserve_message) {
                    echo $reserve_message;
                }
            ?>

            <div class="search-section">
                <!-- search form -->
                <form method="post" class="search-form">
                    <div class="form-group">
                        <label for="title" class="form-label">Book Title:</label>
                        <input type="text" id="title" name="title" class="form-
input"
                                placeholder="Enter full or partial title"
                                value="<?php echo htmlentities($_POST['title']) ??"
                                '); ?>">
                    </div>

                    <div class="form-group">
                        <label for="author" class="form-label">Author:</label>
                        <input type="text" id="author" name="author" class="form-
input"
                                placeholder="Enter full or partial author name"
                                value="<?php echo htmlentities($_POST['author']) ??"
                                '); ?>">
                    </div>

                    <!-- drop down menu for categories -->

```

```

        <div class="form-group">
            <label for="category" class="form-
label">Category:</label>
            <select id="category" name="category" class="form-
select">
                <option value="">All Categories</option>
                <?php foreach($categories as $cat) {?>
                    <option value="<?php echo $cat['id']; ?>"
                        <?php echo (isset($_POST['category']) &&
$_POST['category'] == $cat['id']) ? 'selected' : ''; ?>>
                        <?php echo htmlentities($cat['description']);
?>
                    </option>
                <?php }?>
            </select>
        </div>

        <div class="form-group">
            <input type="submit" value="Search" class="submit-btn">
            <button type="reset" class="reset-btn">Clear</button>
        </div>
    </form>
</div>

<!-- search message -->
<?php echo $search_message; ?>

<!-- output books -->
<?php if(!empty($search_results)) {?>
    <div class="results-section">
        <h2 class="results-title">Search Results (<?php echo
count($search_results); ?> found)</h2>

        <div class="results-grid">
            <?php foreach($search_results as $book) {?>
                <div class="book-card <?php echo $book['reserved'] ?
'reserved' : 'available'; ?>">
                    <div class="book-header">
                        <h3 class="book-title"><?php echo
htmlentities($book['title']); ?></h3>
                        <div class="book-status">
                            <span class="status-badge <?php echo
$book['reserved'] ? 'status-reserved' : 'status-available'; ?>">
                                <?php echo $book['reserved'] ?
'Reserved' : 'Available'; ?>

```

```

        </span>
    </div>
</div>

    <div class="book-details">
        <p class="book-
author"><strong>Author:</strong> <?php echo htmlentities($book['author']); ?></p>
        <p class="book-
category"><strong>Category:</strong> <?php echo
htmlentities($book['category_name']); ?></p>
        <p class="book-isbn"><strong>ISBN:</strong>
<?php echo htmlentities($book['isbn']); ?></p>
        <p class="book-year"><strong>Year:</strong>
<?php echo $book['year']; ?></p>
        <p class="book-
edition"><strong>Edition:</strong> <?php echo $book['edition']; ?></p>
    </div>

    <!-- form to reserve the book if it is available
-->

    <?php if(!$book['reserved']) {?>
        <form method="post" class="reserve-form">
            <input type="hidden" name="reserve_isbn"
value="<?php echo $book['isbn']; ?>">
            <input type="hidden" name="title"
value="<?php echo htmlentities($_POST['title'] ?? ''); ?>">
            <input type="hidden" name="author"
value="<?php echo htmlentities($_POST['author'] ?? ''); ?>">
            <input type="hidden" name="category"
value="<?php echo htmlentities($_POST['category'] ?? ''); ?>">
            <button type="submit" class="reserve-
btn">Reserve This Book</button>
        </form>
        <?php } else {?>
            <div class="reserved-info">
                <p class="reserved-text">This book is
already reserved</p>
            </div>
            <?php }?>
        </div>
    <?php }?>
</div>

    <?php } elseif($_SERVER["REQUEST_METHOD"] == "POST" &&
empty($search_message)) {?>

```

```

        <div class="no-results-message">
            <p>No books found matching your search criteria.</p>
        </div>
    <?php }?>
</div>
</main>

    <?php include 'footer.php'; ?>
</body>
</html>

```

13. Style.css

```

/* basic styling */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    line-height: 1.5;
    color: #333;
    background: #f5f5f5;
}

button {
    transition: background-color 0.3s;
}

/* header style */
.site-header {
    background: #2c3e50;
    color: white;
    padding: 15px 0;
}

.header-container {
    max-width: 1200px;
    margin: 0 auto;
    padding: 0 20px;
    display: flex;

```

```
    justify-content: space-between;
    align-items: center;
}

.site-title {
    font-size: 1.5rem;
}

.nav-list {
    display: flex;
    list-style: none;
    gap: 15px;
}

.nav-link {
    color: white;
    text-decoration: none;
    padding: 5px 10px;
}

.nav-link:hover {
    background: rgba(255,255,255,0.1);
    border-radius: 3px;
}

/* authorization pages style */
.auth-body {
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    padding: 20px;
}

.auth-form-container {
    background: white;
    padding: 30px;
    border-radius: 8px;
    box-shadow: 0 5px 15px rgba(0,0,0,0.1);
    width: 100%;
    max-width: 400px;
}

.auth-title {
    text-align: center;
```

```
    margin-bottom: 20px;
    color: #2c3e50;
}

/* forms styling */
.form-group {
    margin-bottom: 15px;
}

.form-label {
    display: block;
    margin-bottom: 5px;
    font-weight: 500;
}

.form-input {
    width: 100%;
    padding: 10px;
    border: 1px solid #ddd;
    border-radius: 4px;
    font-size: 16px;
}

.form-input:focus {
    outline: none;
    border-color: #3498db;
}

.submit-btn {
    width: 100%;
    padding: 12px;
    background: #3498db;
    color: white;
    border: none;
    border-radius: 4px;
    font-size: 16px;
    cursor: pointer;
    margin-top: 10px;
    transition: background-color 0.3s;
}

.submit-btn:hover {
    background: #2980b9;
}
```

```
/* messages */
.error-message {
  background: #f8d7da;
  color: #721c24;
  padding: 10px;
  border-radius: 4px;
  margin-bottom: 15px;
}

.success-message {
  background: #d4edda;
  color: #155724;
  padding: 10px;
  border-radius: 4px;
  margin-bottom: 15px;
}

/* main content on pages */
.main-content {
  padding: 30px 0;
  min-height: calc(100vh - 130px);
}

.content-container {
  max-width: 1200px;
  margin: 0 auto;
  padding: 0 20px;
}

.page-title {
  font-size: 1.8rem;
  margin-bottom: 20px;
  color: #2c3e50;
}

/* search button */
.search-button {
  display: inline-block;
  background-color: #3498db;
  border: none;
  border-radius: 4px;
  padding: 10px 20px;
  cursor: pointer;
  transition: background-color 0.3s;
}
```



```
.search-button:hover {
  background-color: #2980b9;
}

.search-link {
  color: white;
  text-decoration: none;
  font-weight: bold;
  font-size: 16px;
}

/* book cards */
.books-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
  gap: 20px;
  margin-top: 20px;
}

.book-card {
  background: white;
  padding: 20px;
  border-radius: 8px;
  box-shadow: 0 2px 5px rgba(0,0,0,0.1);
  border-left: 4px solid #3498db;
}

.book-card.reserved {
  border-left-color: #e74c3c;
}

.book-card.available {
  border-left-color: #2ecc71;
}

.book-title {
  font-size: 1.2rem;
  margin-bottom: 10px;
  color: #2c3e50;
}

.book-details p {
  margin: 5px 0;
  color: #666;
}
```

```
}

.reserve-btn {
  width: 100%;
  padding: 10px;
  background: #2ecc71;
  color: white;
  border: none;
  border-radius: 4px;
  margin-top: 10px;
  cursor: pointer;
}

.reserve-btn:hover {
  background: #27ae60;
}

/* tables */
.data-table {
  width: 100%;
  background: white;
  border-radius: 8px;
  overflow: hidden;
  box-shadow: 0 2px 5px rgba(0,0,0,0.1);
}

.table-header {
  background: #34495e;
  color: white;
}

.table-cell {
  padding: 12px 15px;
  text-align: left;
  border-bottom: 1px solid #eee;
}

.table-row:last-child .table-cell {
  border-bottom: none;
}

.remove-btn {
  padding: 6px 12px;
  background: #e74c3c;
  color: white;
}
```

```
border: none;
border-radius: 3px;
cursor: pointer;
}

.remove-btn:hover {
  background: #c0392b;
}

/* index/home page */
.welcome-section {
  background: #3498db;
  color: white;
  padding: 30px;
  border-radius: 8px;
  margin-bottom: 30px;
  text-align: center;
}

.actions-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 20px;
  margin: 20px 0;
}

.action-card {
  background: white;
  padding: 20px;
  border-radius: 8px;
  text-decoration: none;
  color: #333;
  box-shadow: 0 2px 5px rgba(0,0,0,0.1);
  text-align: center;
}

.action-card:hover {
  transform: translateY(-2px);
  box-shadow: 0 4px 10px rgba(0,0,0,0.15);
}

/* footer */
.site-footer {
  background: #2c3e50;
  color: white;
```

```
padding: 20px 0;
text-align: center;
}

/* responsive */
@media (max-width: 768px) {
  .header-container {
    flex-direction: column;
    gap: 10px;
  }

  .nav-list {
    flex-wrap: wrap;
    justify-content: center;
  }

  .books-grid {
    grid-template-columns: 1fr;
  }

  .actions-grid {
    grid-template-columns: 1fr;
  }

  .data-table {
    font-size: 14px;
  }
}
```